MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, DETAIL LIGHTING FOR BERTHS

8 WATTS, 120 VOLTS, 60 HERTZ, SYMBOLS 229.1, 231.1 AND 232.1

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

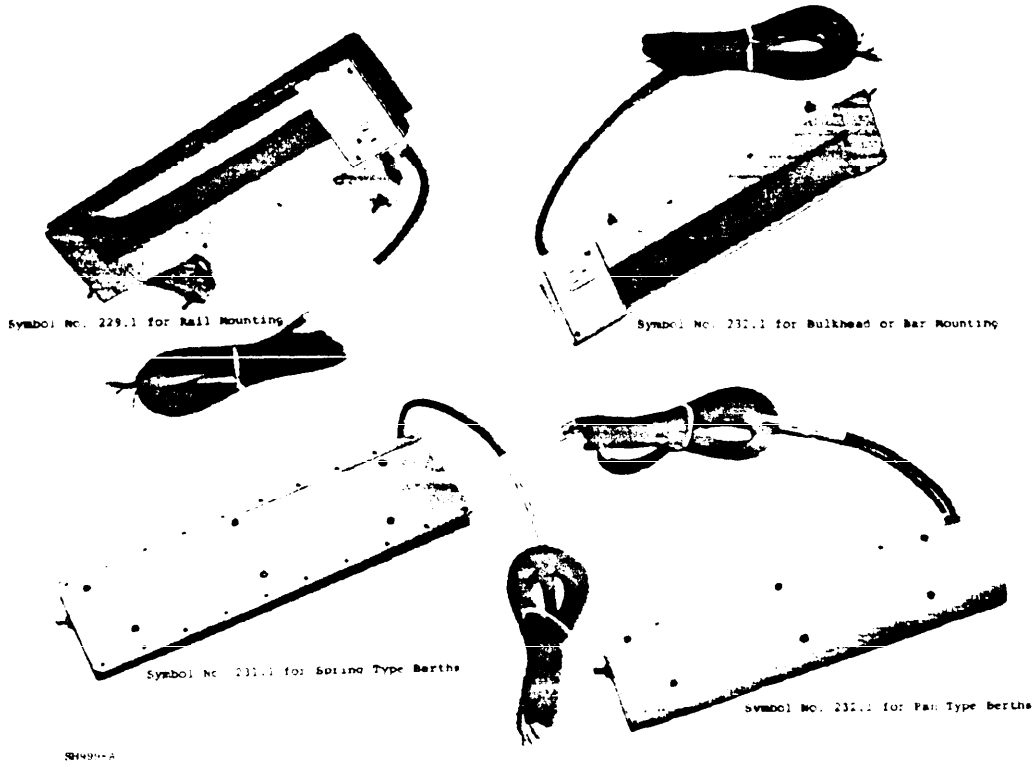
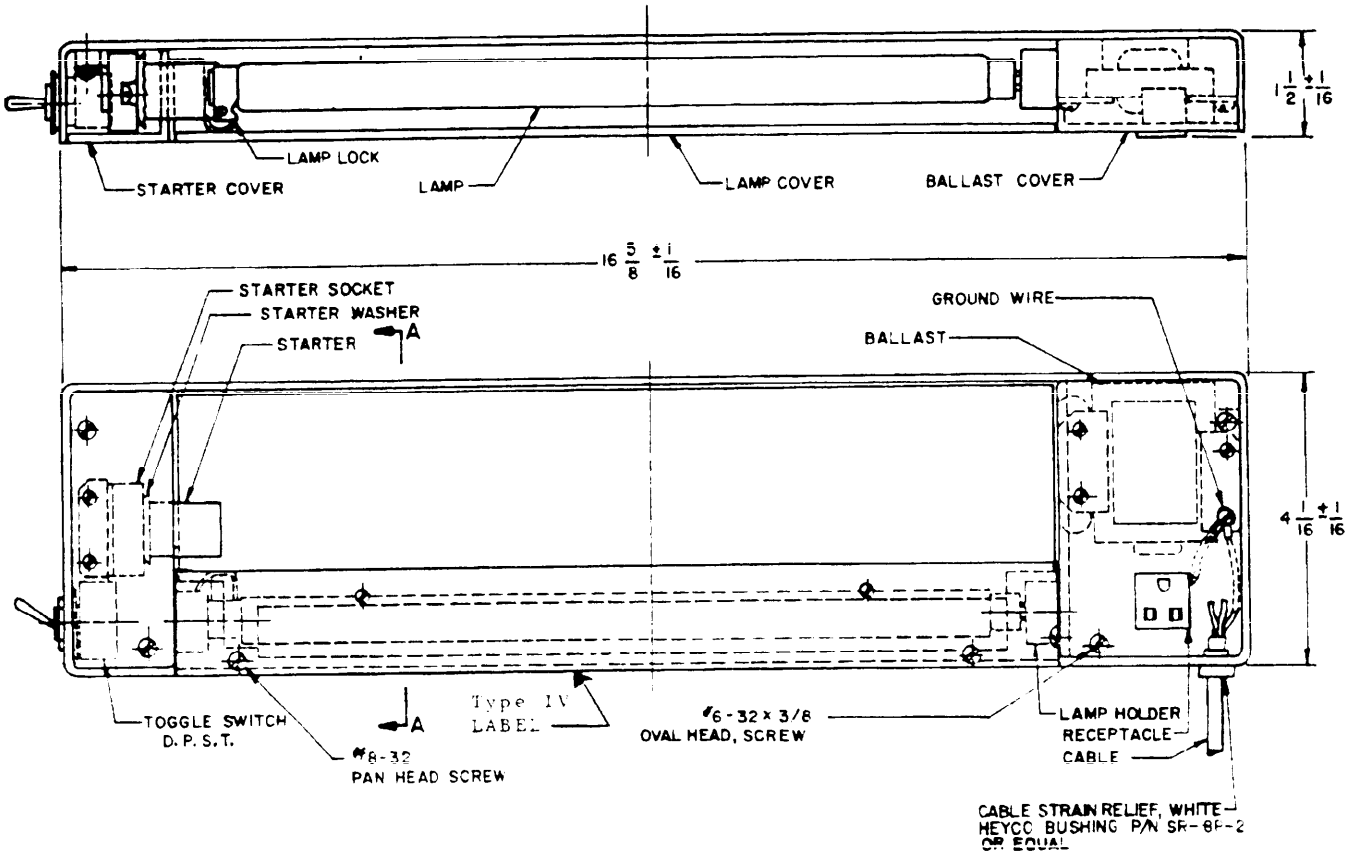


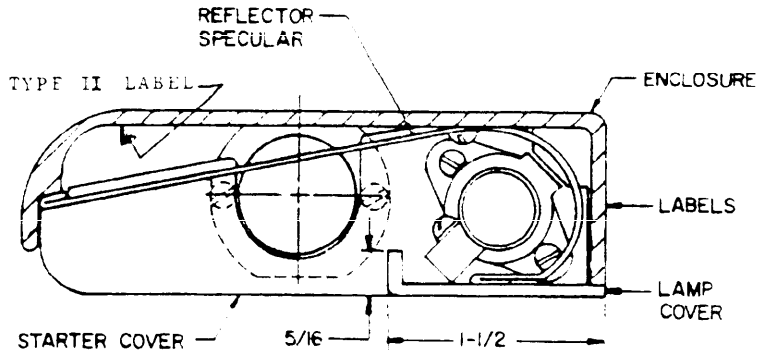
FIGURE 1. Typical view of Symbols 229.1, 231.1 and 232.1.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6230

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SECTION A-A

SH 8671

FIGURE 2. Typical plan and section views of basic light for Symbols 229.1, 231.1 and 232.1.

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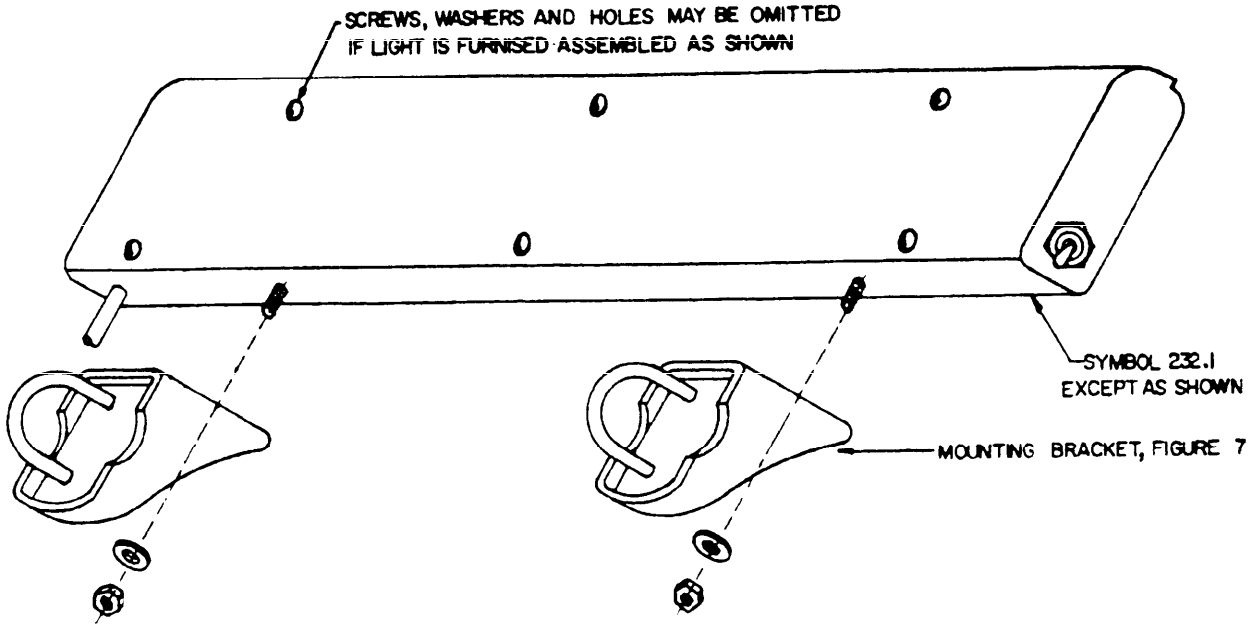


FIGURE 3. Typical view of Symbol 229.1.

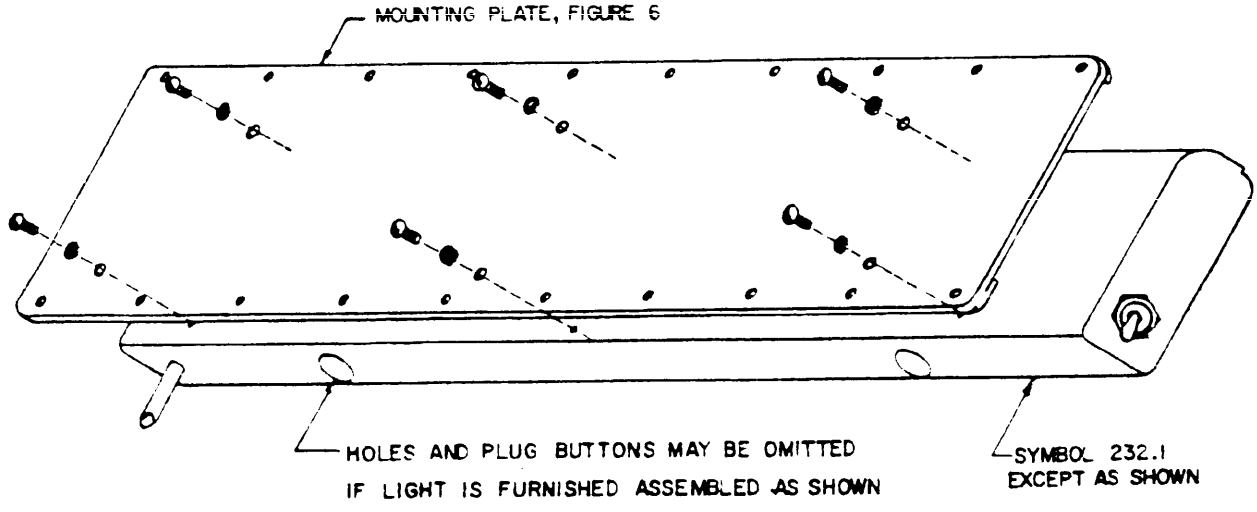


FIGURE 4. Typical view of Symbol 231.1.

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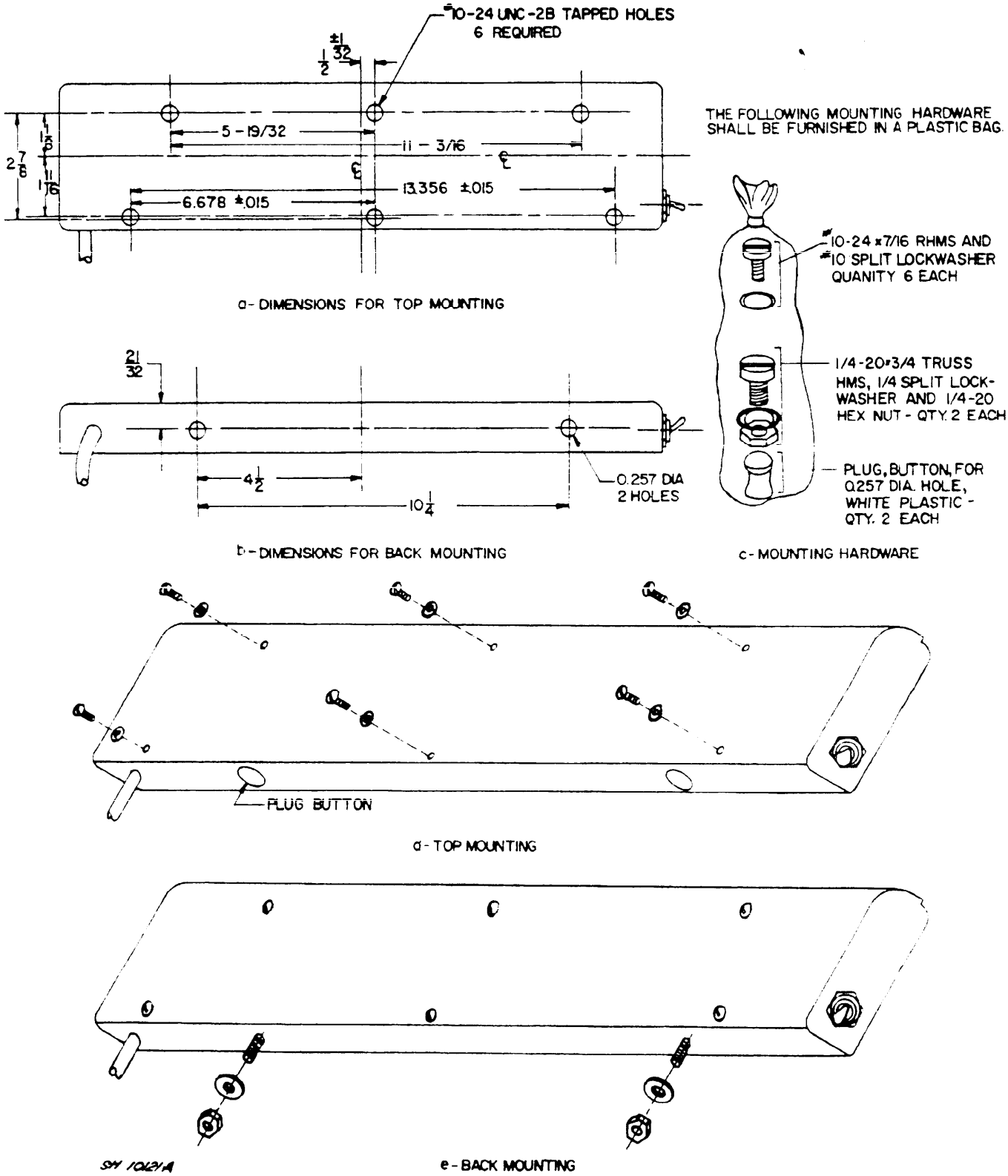


FIGURE 5. Typical view of Symbol 232.1.

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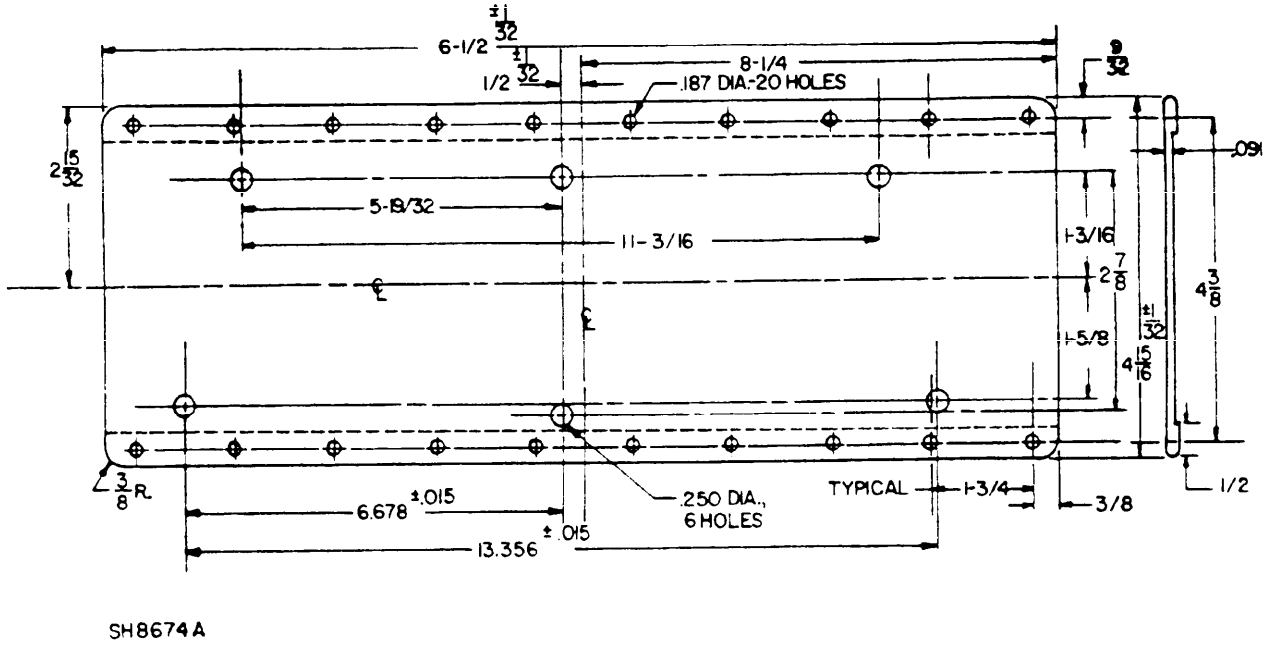


FIGURE 6. Mounting plate for Symbol 231.1.

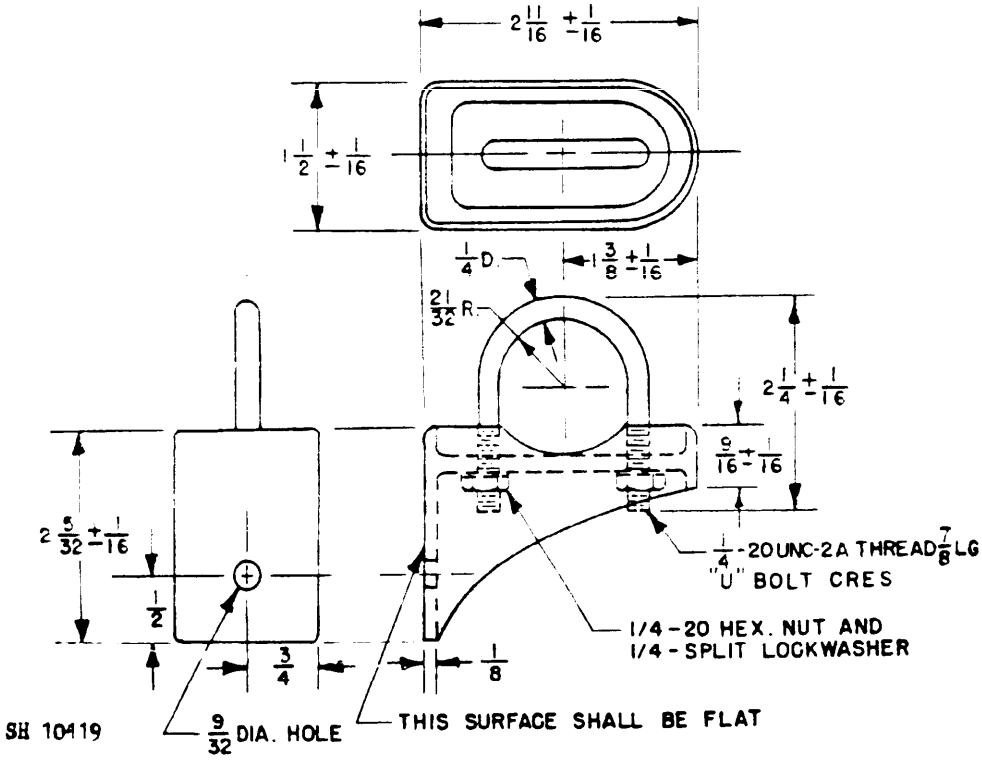


FIGURE 7. Mounting bracket for Symbol 229.1.

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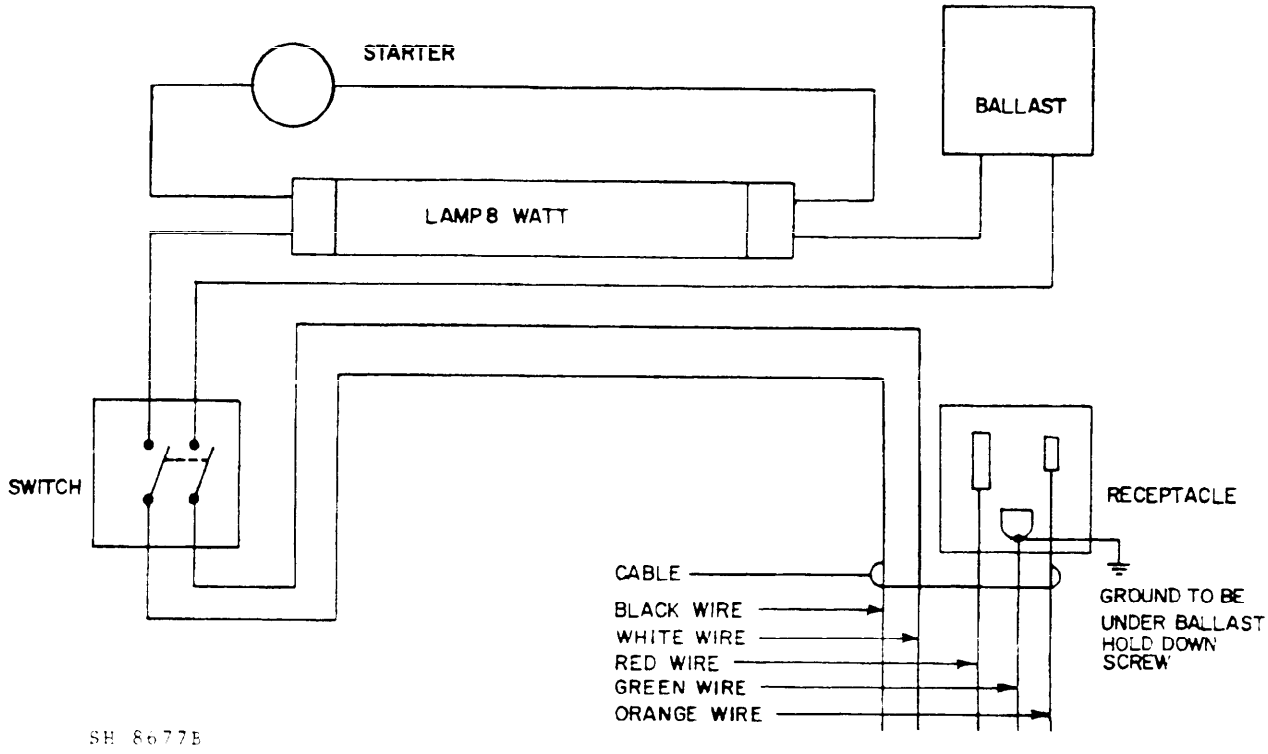
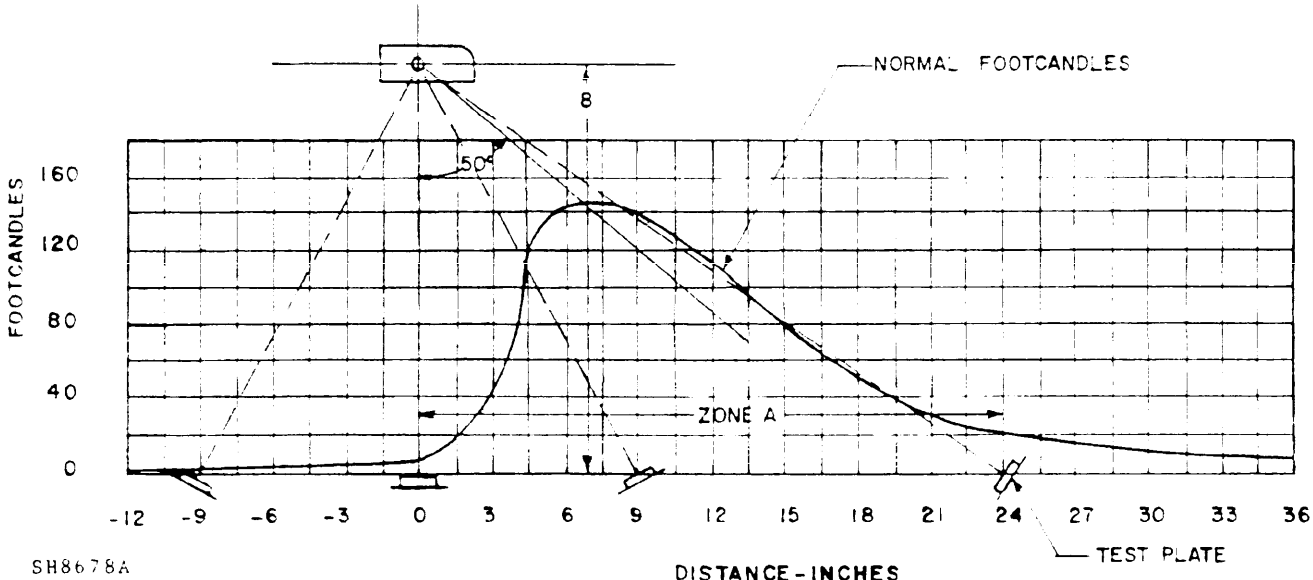


FIGURE 8. Wiring diagram.



Photometric test data:

Distance: 8 inches.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamp: Lamp furnished with light shall be used during test.
Ballast: Ballast furnished with light shall be used during test.
Lamp correction: Distribution curves based on 380 lumens per lamp.

FOOTCANDLE VALUES WITHIN ZONE "A" SHALL BE NOT LESS THAN 90 PERCENT OF THE VALUES SHOWN HEREON. THESE RESULTS SHALL BE MET AFTER CORRECTION TO 380 LUMENS OUTPUT FOR BOTH LAMP AND BALLAST HAS BEEN MADE.

FIGURE 9. Footcandle distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: 1
Class: 1

SYMBOL NUMBERS:

229.1: Complete lighting fixture assembly, for rail mounting, with one 8-watt cool white fluorescent lamp.
231.1: Complete lighting fixture assembly, for spring type berths (bunks), with one 8-watt cool white fluorescent lamp.
232.1: Complete lighting fixture assembly, for pan type berths bulkhead or bar mounting, with one 8-watt cool white fluorescent lamp.

DESIGN:
Electrical rating: 8 watts, 120 volts, 60 Hertz.
Dimensions and configurations: See figures 1 through 7.
Photmetric data: See figure 9.
Lampholder spacing: See figure 21 of MIL-F-16377.
Wiring diagram: See figure 8.
Material: Material for enclosure, lamp cover, mounting plate and mounting brackets shall be aluminum or plastic type MAI-60, in accordance with MIL-M-14.
Finish: Finish shall comply with 3.8.3.4 of MIL-F-16377, except that color shall be beige conforming to Color Number 26586 of FED-STD-595.
Cable strain relief: Strain relief for cable shall be provided as shown on figure 2.
Grounding: Aluminum enclosure: See figure 2; Plastic enclosure: The reflector, ballast frame and exterior metal parts of switch shall be grounded to receptacle ground contact.
Receptacle: Receptacle shall be Leviton part number 1374-051 (ivory) or equivalent except that the ground contact shall be nickel-plated 0.0003 inch thick.

Cable: 10 feet of type ST, AWG size 18/5, five conductor in accordance with J-C-580 with light gray thermoplastic jacket.
Weight: 4 lbs. 2 oz. maximum
Reflector: Reflector shall be made of specular, pre-finished lighting sheet.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figure 2.

(a) One type I.
(b) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I by name and stock number only.
(c) One type IV with the information "RECEPTACLE WIRES (RED AND ORANGE) SHALL BE CONNECTED TO ISOLATED RECEPTACLE CIRCUIT ONLY. OTHER WIRES SHALL NOT BE CONNECTED TO ISOLATED RECEPTACLE CIRCUIT."

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE:

First article inspection:
Complete fixtures: See table II.

Quality conformance: Same as the first article inspection.

Inspection of product for delivery:
Examination: See MIL-F-16377.
Examination for preparation of delivery: MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table III.

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TABLE I. Standard parts.

Name	Specification or Military part number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry Number F8T5/CW
Lamp lock	M16377/40-001	6250-00-399-7039	
Lampholder	MIL-L-970/10	6250-00-377-8590	
Startersocket	MIL-L-970/23	6250-00-295-2738	
Starter	8W, Type III of W-S-755	6250-00-299-5962	Commercial Designation: FS5
Ballast	M16377/45-001	6250-00-295-2739	
Switch, toggle, DPST	MS16569-1	5930-00-719-9659	

TABLE II. First article and comparison inspection.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Dielectric	3.6.15	4.8.2	
Insulation Resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Grounding circuit	3.6.13.2	4.8.11	
Light output	3.7.2	4.8.6.2	Footcandle
Magnetic permeability	3.4.8	4.8.18	

TABLE III. Part number and general information.

Military Part Number M16377/17-	Item name	National Stock Number	Remarks
229.1	Complete light, Symbol 229.1		1/ 2/ Fig. 3
231.1	Complete light, Symbol 231.1		1/ 2/ Fig. 4
232.1	Complete light, Symbol 232.1		1/ 2/ Fig. 5
001	Mounting bracket for Symbol 229.1 -two brackets		1/ Fig. 7
002	Mounting plate for Symbol 231.1		1/ Fig. 6

1/ Formerly covered on NAVSHIPS Drawing 815-1197113 and MS 17114. Symbol 234 has been superseded by Symbol 232. Symbol 230 (M16377/17-230) has been superseded by the C.G. Doane Co. Part Number 430 or equivalent.

2/ Former berth lights without receptacle (Symbol 229-NSN6230-00-578-7547, Symbol 231-NSN6230-00-808-4466 and Symbol 232-NSN6230-00-578-7458), if need to be procured for replacement purposes for ships where isolated receptacle circuits have not been provided for berth lights, shall be in accordance with symbols 229.1, 231.1 and 232.1 respectively as specified herein except that the receptacle shall not be furnished and the cable shall be a three conductor.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6230-N264)

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